POLYNOMIAL CONFIGURATIONS AND POINTWISE AVERAGES ALONG PIATETSKI–SHAPIRO SEQUENCES

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ABSTRACT. In this paper, we prove that for every integer $k \geq 2$ and every $c > 1$ sufficiently close to 1, there is $\kappa > 0$ such that every sufficiently large subset of $\{1, \dots, N\}$ of density at least $(\log\log N)^{-\kappa}$ contains

$$x, \quad x + \lfloor n^c \rfloor, \quad x + \lfloor n^c \rfloor^2, \quad \dots, \quad x + \lfloor n^c \rfloor^k.$$

We also prove pointwise almost-everywhere convergence of the associated multiple ergodic averages.

1. INTRODUCTION

Bergelson and Leibman proved the polynomial Szemerédi theorem [1], which states that every set of integers of positive upper density contains polynomial configurations with a common parameter. Peluse subsequently obtained the first quantitative result for general distinct-degree families, proving that every sufficiently large subset of $[N]$ with density at least $(\log\log N)^{-\kappa}$ contains polynomial configurations [12]. Shao and Wang recently strengthened this to a polylogarithmic density bound, while also proving a quantitative popular-difference theorem [16, Theorems 1.1 and 1.2].

Sárközy established one of the first results where the common difference belongs to a sparse subset of integers with special structures. Indeed, he proved that every set of integers with positive density contains a two-term configuration with a square or shifted prime difference [14, 15]. In this direction, Piatetski–Shapiro sequences have been studied both as restricted sets of common differences and as sparse sampling sequences in ergodic theory. Frantzikinakis and Wierdl proved that every set of positive upper density contains arbitrarily long arithmetic progressions whose common difference is of the form $\lfloor n^c \rfloor$ [7, Theorem A]. For the configuration $x, x + \lfloor n^c \rfloor$, Rivat and Sárközy obtained a power-saving density bound [13], which was recently strengthened for $1 < c < 6/5$ by O'Keeffe [11, Theorem 1.3]. Boshernitzan and Wierdl proved pointwise convergence for single ergodic averages along $\lfloor n^c \rfloor$ [2]. More recently, Daskalakis established bilinear ergodic and Roth-type results along fractional powers, including configurations of the form $\{x, x + \lfloor n^c \rfloor, x + 2\lfloor n^c \rfloor\}$ for $c > 1$ sufficiently close to 1 [4, Theorem 1.7]. These results concern either a single difference or equal-degree linear configurations. In this paper, we establish a quantitative distinct-degree polynomial Szemerédi theorem along Piatetski–Shapiro integers and prove the corresponding pointwise multiple ergodic theorem.

For $c > 1$, write

$$\mathcal{S}_c := \{\lfloor n^c \rfloor : n \in \mathbb{N}\}.$$

Fix throughout

$$P_0 = 0, \qquad P_1, \dots, P_\ell \in \mathbb{Z}[y], \qquad P_i(0) = 0, \qquad 1 \leq \deg P_1 < \cdots < \deg P_\ell = d. \tag{1}$$

The vanishing constant-term assumption is standard in polynomial recurrence. Our first result is a quantitative nonlinear Szemerédi theorem along $\mathcal{S}_c$. (Note that we do not recover the polylogarithmic threshold of [16]: our comparison requires their popular-difference estimate, whose quantitative dependence yields the double-logarithmic bound above.)

Theorem 1.1 (Piatetski–Shapiro polynomial Szemerédi theorem)**.** *For every family satisfying* (1)*, there are $c_* > 1$ and $\kappa > 0$ such that the following holds. If $1 < c < c_*$, then every sufficiently large $A \subset [N]$ satisfying*

$$|A| \geq \frac{N}{(\log\log N)^\kappa}$$

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contains

$$x, \quad x + P_1(\lfloor n^c \rfloor), \quad \dots, \quad x + P_\ell(\lfloor n^c \rfloor) \tag{2}$$

for some positive integers x, n. Moreover, x, n may be chosen so that the displayed terms are pairwise distinct.

The argument also gives the following quantitative counting estimate, based on the popular-difference theorem of Shao–Wang. For $A \subset [N]$ and $M = \lfloor N^{1/d} \rfloor$, define

$$\mathcal{C}_{c,\mathbf{P}}(A; M) := \big|\big\{(x, n) \in \mathbb{Z} \times \mathbb{N} : \lfloor n^c \rfloor \le M,\ x + P_i(\lfloor n^c \rfloor) \in A \quad (0 \le i \le \ell)\big\}\big|.$$

For every $1 < c < c_*$, there are constants $C, \zeta > 0$, depending only on the fixed polynomial family, such that, writing $0 < \delta = |A|/N$,

$$\frac{\mathcal{C}_{c,\mathbf{P}}(A; M)}{NM^{1/c}} \ge \exp\big(-\exp(C\delta^{-C})\big) - O_{c,\mathbf{P}}(M^{-\zeta}) \tag{3}$$

whenever $N \ge \exp\big(\exp(C\delta^{-C})\big)$.

The connection between Szemerédi-type theorems and multiple recurrence was studied by Furstenberg [9] and was extended to polynomial configurations by Bergelson and Leibman [1]. This connection suggests a corresponding ergodic theorem analogous to Theorem 1.1. For Piatetski–Shapiro times, Frantzikinakis [8, Theorem 2.1] proved norm convergence for linear multiple averages. Daskalakis proved that single averages along orbits $\lfloor n^c \rfloor^2$ converge pointwise to the ordinary quadratic limit [4, Theorem 1.13]. In separate work, he proved pointwise convergence for symmetric double recurrence averages along $\mathcal{S}_c$ when $1 < c < 23/22$ [5, Theorem 1.7]. Fornal and Krause subsequently extended this range to $1 < c < 7/6$, and also obtained a weighted version [6, Proposition 1.5 and Remark 4]. In this paper, we prove pointwise convergence for the general distinct-degree polynomial orbits considered here.

Let (X, ν, T) be an invertible probability measure-preserving system and let $f_1, \dots, f_\ell \in L^\infty(X)$. Put

$$A_R^{(c)}(\mathbf{f})(x) := \frac{1}{R} \sum_{n \le R} \prod_{i=1}^{\ell} f_i\big(T^{P_i(\lfloor n^c \rfloor)} x\big). \tag{4}$$

Theorem 1.2 (Pointwise Piatetski–Shapiro averages)**.** *There exists $c_* > 1$ such that the following statement holds. Given $1 < c < c_*$, the averages* (4) *converge almost everywhere, and their limit agrees almost everywhere with the limit of*

$$\frac{1}{M} \sum_{m \le M} \prod_{i=1}^{\ell} f_i\big(T^{P_i(m)} x\big). \tag{5}$$

The usual PET induction bounds a weighted polynomial count by a Gowers norm. The inverse theorem then shows that a large polynomial count forces correlation with a nilsequence. In the case of averages along $\mathcal{S}_c$, controlling all such obstructions is unnecessary. The appropriate input is the following distinct-degree generalized von Neumann theorem of Teräväinen [17, Theorem 4.1], which identifies ordinary polynomial phases as sufficient obstructions.

Write $[N] = \{1, \dots, N\}$, $\mathbf{P} = (P_1, \dots, P_\ell)$, $\mathrm{e}(t) := e^{2\pi i t}$, $\sum_{m \le M} f(m) = \sum_{0 < m \le M} f(m)$, and $\mathbb{E}_{m \le M} = M^{-1} \sum_{m \le M}$. Unrestricted sums over x are over $\mathbb{Z}$. A function is 1-bounded if its absolute value is at most 1. For $w : [M] \to \mathbb{C}$, define

$$\|w\|_{u^s[M]} := \sup_{\substack{\phi \in \mathbb{R}[y] \\ \deg \phi < s}} \left|\mathbb{E}_{m \le M} w(m) \mathrm{e}(-\phi(m))\right|. \tag{6}$$

Theorem 1.3 ([17])**.** *Let $Q_1, \dots, Q_\ell \in \mathbb{Z}[y]$ satisfy $\deg Q_1 < \cdots < \deg Q_\ell = d$, and let $C_0 \ge 1$. There is an integer $1 \le K \ll_d 1$ such that the following holds. If $F_0, \dots, F_\ell : \mathbb{Z} \to \mathbb{C}$ are 1-bounded and supported on $[-C_0 M^d, C_0 M^d]$, and $\theta : [M] \to \mathbb{C}$ is 1-bounded, then*

$$\left| \frac{1}{M^{d+1}} \sum_{x \in \mathbb{Z}} \sum_{m \le M} \theta(m) F_0(x) \prod_{i=1}^{\ell} F_i(x + Q_i(m)) \right| \ll_{C_0, \mathbf{Q}} \left(M^{-1} + \|\theta\|_{u^{d+1}[M]}\right)^{1/K}. \tag{7}$$

We shall use the following immediate scaled form. If $|g| \le B$ on $[M]$ and

$$\sup_{\deg \phi \le d} |\mathbb{E}_{m \le M} g(m) \mathrm{e}(\phi(m))| \le E, \tag{8}$$

then, for $B > 0$, the left-hand side of (7), with g in place of θ, is

$$\ll_{C_0,\mathbf{Q}} B\big(M^{-1} + B^{-1}E\big)^{1/K}. \tag{9}$$

Indeed, apply Theorem 1.3 to g/B.

We use the estimate in (8) to approximate the contribution from $\mathbf{1}_{\mathcal{S}_c}$ with a more tractable weighted counting form. Vaaler's approximation reduces the problem to exponential sums with phase

$$Q(m) + hm^{1/c}, \qquad \deg Q \le d.$$

Note that the $(d+1)$-st derivative removes Q. Combining this power-saving comparison with the quantitative counting result of Shao and Wang [16, Proposition 5.1 and Lemma 5.2] proves Theorem 1.1. Its orbit version, together with the pointwise theorem of Kosz–Mirek–Peluse–Wan–Wright [10, Theorem 1.3], proves Theorem 1.2.

For the ergodic application we record the corresponding orbit estimate.

Proposition 1.4. *Let (X, ν, T) be an invertible probability measure-preserving system, and let $Q_1, \dots, Q_\ell$ be as in Theorem 1.3. Suppose that $B > 0$, that $|g| \le B$ on $[M]$, and that g satisfies* (8). *Then*

$$\left\| \mathbb{E}_{m \le M} g(m) \prod_{i=1}^{\ell} f_i(T^{Q_i(m)}\cdot) \right\|_{L^1(X)} \ll_{\mathbf{Q}} B\big(M^{-1} + B^{-1}E\big)^{1/K} \prod_{i=1}^{\ell} \|f_i\|_\infty. \tag{10}$$

Proof. By homogeneity assume $\|f_i\|_\infty \le 1$. Since the class of polynomial phases of degree at most d is closed under negation, (8) gives

$$\|g\|_{u^{d+1}[M]} \le E.$$

Put $L = M^d$. By L^1–L^∞ duality, there is $h \in L^\infty(X)$ with $\|h\|_\infty \le 1$ such that

$$\left\| \mathbb{E}_{m \le M} g(m) \prod_{i=1}^{\ell} f_i(T^{Q_i(m)}\cdot) \right\|_{L^1(X)} = \left| \int_X h(x) \mathbb{E}_{m \le M} g(m) \prod_{i=1}^{\ell} f_i(T^{Q_i(m)}x)\, d\nu(x) \right|.$$

Since ν is T-invariant, averaging the integral over $1 \le a \le L$ gives

$$\begin{aligned} &\left\| \mathbb{E}_{m \le M} g(m) \prod_{i=1}^{\ell} f_i(T^{Q_i(m)}\cdot) \right\|_{L^1(X)} \\ &\quad = \left| \int_X \frac{1}{L} \sum_{a \le L} h(T^a x) \mathbb{E}_{m \le M} g(m) \prod_{i=1}^{\ell} f_i(T^{a+Q_i(m)}x)\, d\nu(x) \right|. \end{aligned}$$

For fixed x, define $F_0(a) = h(T^a x)\mathbf{1}_{[L]}(a)$ and define $F_i(b) = f_i(T^b x)$ on an interval containing all $a + Q_i(m)$ with $a \in [L]$ and $m \in [M]$, extending by zero elsewhere. These intervals may be chosen inside $[-C_0 M^d, C_0 M^d]$ for some $C_0 = C_0(\mathbf{Q})$. Thus the functions are 1-bounded and have the support required in Theorem 1.3. Since $L = M^d$, (9) gives, uniformly in x,

$$\left| \frac{1}{L} \sum_{a \le L} h(T^a x) \mathbb{E}_{m \le M} g(m) \prod_{i=1}^{\ell} f_i(T^{a+Q_i(m)}x) \right| \ll_{C_0,\mathbf{Q}} B\big(M^{-1} + B^{-1}E\big)^{1/K}.$$

Integrating this estimate over X proves (10). □

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2. A nonlinear Szemerédi theorem

Put $\gamma = 1/c$. The map $n \mapsto \lfloor n^c \rfloor$ is strictly increasing, since $(n+1)^c - n^c > 1$. Define

$$\chi_c(m) := \mathbf{1}_{\mathcal{S}_c}(m), \qquad \Delta_c(m) := (m+1)^\gamma - m^\gamma, \qquad r_c(m) := \chi_c(m) - \Delta_c(m). \tag{11}$$

The exact counting identity

$$\chi_c(m) = \lceil (m+1)^\gamma \rceil - \lceil m^\gamma \rceil$$

and $\lceil t \rceil = t + \psi(-t) + 1/2$, where $\psi(t) = \{t\} - 1/2$, yield

$$r_c(m) = \psi(-(m+1)^\gamma) - \psi(-m^\gamma). \tag{12}$$

We shall also use

$$0 < \Delta_c(m) < 1, \qquad |r_c(m)| \le 1, \qquad \sum_{m \le M} \Delta_c(m) = (M+1)^\gamma - 1. \tag{13}$$

We require two standard exponential-sum tools. The first is the following consequence of Vaaler's approximation to the sawtooth function [18, Theorem 18, especially (7.14)]. We include a short proof for completeness.

Lemma 2.1 (Vaaler approximation)**.** *For every integer $H \ge 2$ there are coefficients a_h, b_h, with $0 < |h| \le H$, and a nonnegative trigonometric polynomial*

$$B_H(t) = b_0 + \sum_{0<|h|\le H} b_h \mathrm{e}(ht),$$

such that

$$|a_h| \ll |h|^{-1}, \qquad b_0 \ll H^{-1}, \qquad |b_h| \ll H^{-1},$$

and

$$\left| \psi(t) - \sum_{0<|h|\le H} a_h \mathrm{e}(ht) \right| \le B_H(t)$$

for every real t.

Proof. Vaaler uses the normalized sawtooth function

$$\widetilde{\psi}(t) = \begin{cases} \{t\} - \frac{1}{2}, & t \notin \mathbb{Z}, \\ 0, & t \in \mathbb{Z}. \end{cases}$$

Taking $N = H$ in [18, Theorem 18], define

$$a_h := (-2\pi i h)^{-1} \widehat{J}_{H+1}(h), \qquad 0 < |h| \le H.$$

Then [18, (7.14)] gives

$$\left| \widetilde{\psi}(t) - \sum_{0<|h|\le H} a_h \mathrm{e}(ht) \right| \le \frac{1}{2H+2} k_H(t),$$

where

$$k_H(t) = \sum_{|h|\le H} \left(1 - \frac{|h|}{H+1}\right) \mathrm{e}(ht)$$

is the Fejér kernel. Set $B_H = (2H+2)^{-1} k_H$. The properties of $\widehat{J}$ in [18, Theorem 6] give $|\widehat{J}_{H+1}(h)| \le 1$, and hence $|a_h| \ll |h|^{-1}$. Moreover,

$$b_h = \frac{1}{2H+2}\left(1 - \frac{|h|}{H+1}\right) \qquad (|h| \le H),$$

so $b_0, |b_h| \ll H^{-1}$. Finally, our convention $\psi(t) = \{t\} - 1/2$ differs from $\widetilde{\psi}$ only on $\mathbb{Z}$. Since $\widehat{J}$ is even, we have $a_{-h} = -a_h$. Thus, if $t \in \mathbb{Z}$, then

$$\sum_{0<|h|\le H} a_h \mathrm{e}(ht) = \sum_{h=1}^{H} (a_h + a_{-h}) = 0.$$

Moreover, $k_H(t) = H+1$, and hence $B_H(t) = 1/2$. Since $\psi(t) = -1/2$ at such points, the required inequality holds there as well. □

The second is the following form of the classical higher derivative test; see [3, Lemma 2.2].

Lemma 2.2 (Higher derivative test)**.** *Let $s \geq 2$, let J be an interval of L integers, and suppose that F is s times continuously differentiable on the convex hull of J. If*

$$\lambda \leq |F^{(s)}(t)| \leq C_1\lambda$$

throughout that interval, then

$$\left|\sum_{m\in J} \mathrm{e}(F(m))\right| \ll_{s,C_1} L\left(\lambda^{1/(2^s-2)} + L^{-2/2^s} + (L^s\lambda)^{-2/2^s}\right). \tag{14}$$

The required arithmetic input is the following uniform estimate for the correlation of the Piatetski–Shapiro sampling discrepancy with polynomial phases.

Lemma 2.3. *Fix $d \geq 1$ and $1 < c_0 < 2$. There are $\theta, \sigma_0 > 0$, depending only on d, c_0, such that the following holds. If $1 < c < c_0$, $\gamma = 1/c$, $Q \in \mathbb{R}[y]$ has degree at most d, and $1 \leq |h| \leq M^\theta$, then*

$$\left|\sum_{m\leq M} \mathrm{e}(Q(m) + hm^\gamma)\right| \ll_{c,d} M^{1-\sigma_0}. \tag{15}$$

The same estimate holds with m^γ replaced by $(m+1)^\gamma$.

Proof. Choose $0 < \theta < d/4$. The range $m \leq M^{1/2}$ contributes $O(M^{1/2})$. Partition the remaining range into dyadic intervals $J \subset [Y, 2Y]$, where $Y \geq M^{1/2}$.

Since the $(d+1)$-st derivative of Q vanishes, for $t \in [Y, 2Y]$ we have

$$\left|\frac{d^{d+1}}{dt^{d+1}}\big(Q(t) + ht^\gamma\big)\right| \asymp_{c,d} |h|Y^{\gamma-d-1}.$$

Lemma 2.2 therefore gives

$$\begin{aligned}\frac{1}{|J|}\left|\sum_{m\in J} \mathrm{e}(Q(m) + hm^\gamma)\right| \ll_{c,d} {}& |h|^{1/(2^{d+1}-2)}Y^{-(d+1-\gamma)/(2^{d+1}-2)} \\ & + Y^{-2/2^{d+1}} + (|h|Y^\gamma)^{-2/2^{d+1}}.\end{aligned}$$

Since $Y \geq M^{1/2}$, $|h| \leq M^\theta$, $d+1-\gamma \geq d$, and $\gamma \geq 1/c_0$, these terms are bounded, respectively, by

$$M^{-(d/2-\theta)/(2^{d+1}-2)}, \qquad M^{-1/2^{d+1}}, \qquad M^{-1/(c_0 2^{d+1})}.$$

Summing over the dyadic intervals and combining this with the initial range, and then choosing $\sigma_0 > 0$ smaller than the exponents obtained above, proves (15).

Finally, after the change of variables $r = m + 1$, the sum with $(m+1)^\gamma$ has polynomial part $Q(r-1)$ and differs from a sum of the form (15) by at most two boundary terms. □

Proposition 2.4 (Polynomial-phase cancellation)**.** *Fix $d \geq 1$ and $1 < c_0 < 2$. There is $\sigma > 0$, depending only on d, c_0, such that, for every fixed $1 < c < c_0$ and all $M \geq M_0(c)$,*

$$\sup_{\deg Q \leq d} |\mathbb{E}_{m\leq M} r_c(m)\mathrm{e}(Q(m))| \ll_{c,d} M^{-\sigma}, \tag{16}$$

where r_c is defined in (11)*.*

Proof. Apply Lemma 2.1 to the two sawtooth functions in (12), with $H = \lfloor M^\theta \rfloor$ and θ as in Lemma 2.3. We obtain

$$\left|r_c(m) - \sum_{1\leq|h|\leq H} a_h\big(\mathrm{e}(-h(m+1)^\gamma) - \mathrm{e}(-hm^\gamma)\big)\right| \leq B_H(-(m+1)^\gamma) + B_H(-m^\gamma).$$

Given $Q \in \mathbb{R}[y]$ with $\deg Q \leq d$, multiply by $\mathrm{e}(Q(m))$ and sum over $m \leq M$. It follows that

$$\begin{aligned}\left|\sum_{m\leq M} r_c(m)\mathrm{e}(Q(m))\right| \leq {}& \left|\sum_{m\leq M} \mathrm{e}(Q(m)) \sum_{1\leq|h|\leq H} a_h\big(\mathrm{e}(-h(m+1)^\gamma) - \mathrm{e}(-hm^\gamma)\big)\right| \\ & + \sum_{m\leq M} \big(B_H(-(m+1)^\gamma) + B_H(-m^\gamma)\big).\end{aligned} \tag{17}$$

By the triangle inequality and $|a_h| \ll |h|^{-1}$, the first term is at most

$$\ll \sum_{1\le|h|\le H} \frac{1}{|h|}\Bigg(\left|\sum_{m\le M} \mathrm{e}(Q(m)-hm^\gamma)\right| + \left|\sum_{m\le M} \mathrm{e}(Q(m)-h(m+1)^\gamma)\right|\Bigg)$$
$$\ll_{c,d} M^{1-\sigma_0} \sum_{1\le|h|\le H} \frac{1}{|h|} \ll_{c,d} M^{1-\sigma_0}\log M,$$

by Lemma 2.3.

For the second term in (17), the Fourier expansion of B_H and the triangle inequality give

$$\begin{aligned}
&\sum_{m\le M} \big(B_H(-(m+1)^\gamma) + B_H(-m^\gamma)\big)\\
&\quad\le 2Mb_0 + \sum_{1\le|h|\le H} |b_h| \left(\left|\sum_{m\le M} \mathrm{e}(-hm^\gamma)\right| + \left|\sum_{m\le M} \mathrm{e}(-h(m+1)^\gamma)\right|\right)\\
&\quad\ll \frac{M}{H} + \frac{1}{H}\sum_{1\le|h|\le H} \left(\left|\sum_{m\le M} \mathrm{e}(-hm^\gamma)\right| + \left|\sum_{m\le M} \mathrm{e}(-h(m+1)^\gamma)\right|\right)\\
&\quad\ll_{c,d} M^{1-\theta} + M^{1-\sigma_0},
\end{aligned}$$

where the last estimate follows from Lemma 2.3 with $Q = 0$. Dividing by M, taking the supremum over Q, and choosing $0 < \sigma < \min\{\theta, \sigma_0\}$ proves the proposition. □

Combining Proposition 2.4 with Theorem 1.3 gives the comparison used in both applications.

Proposition 2.5 (Piatetski–Shapiro transference)**.** *There are $\beta > 0$ and $c_* > 1$, depending only on the family* (1)*, such that, whenever $1 < c < c_*$ and $F_0, \dots, F_\ell$ are 1-bounded and supported on a fixed multiple of $[-M^d, M^d]$,*

$$\left|\frac{1}{M^{d+1}}\sum_x \sum_{m\le M} r_c(m) \prod_{i=0}^{\ell} F_i(x + P_i(m))\right| \ll_{c,\mathbf{P}} M^{-\beta}. \tag{18}$$

The same constants may be chosen so that, writing $\gamma = 1/c$,

$$1 - \gamma < \beta/2 \tag{19}$$

and

$$\left\|\mathbb{E}_{m\le M} r_c(m) \prod_{i=1}^{\ell} f_i(T^{P_i(m)}\cdot)\right\|_1 \ll_{c,\mathbf{P}} M^{-\beta} \prod_i \|f_i\|_\infty. \tag{20}$$

Proof. Use (16) in Theorem 1.3 and Proposition 1.4, with $B = 1$. Both right-hand sides are $O(M^{-\beta})$ for some $\beta > 0$ depending only on the fixed polynomial family and a fixed upper bound $c_0 > 1$. The exponent is uniform for $1 < c < c_0$; only the implied constant and the threshold for M may depend on c. Finally shrink c_* so that $1 - 1/c < \beta/2$ for $1 < c < c_*$. □

We recall the local-factor terminology from [16, Definitions 3.1, 3.2, 3.4 and 3.6]. A factor $\mathcal{B}$ of $[N]$ is a partition of $[N]$. If $\mathcal{B}(x)$ denotes the atom containing x, define

$$(\Pi_{\mathcal{B}} f)(x) := \frac{1}{|\mathcal{B}(x)|}\sum_{y\in\mathcal{B}(x)} f(y),$$

extended by zero outside $[N]$. The factor $\mathcal{B}$ is local of modulus q and resolution M if every atom is an arithmetic progression of step q and cardinality between M and $2M$. A local factor chain $\mathcal{B}_1, \dots, \mathcal{B}_\ell$ of common modulus q is a sequence of local factors such that $\mathcal{B}_i$ refines $\mathcal{B}_{i+1}$ for every $i < \ell$.

The remaining combinatorial input is the following consequence of Shao–Wang's quantitative popular-difference [16, Theorem 1.2].

Lemma 2.6. *There is $C = C(\mathbf{P}) \geq 1$ such that the following holds. Let $A \subset [N]$ have density δ, and put $M_0 = \lfloor N^{1/d} \rfloor$. If*

$$N \geq \exp\bigl(\exp(C\delta^{-C})\bigr), \tag{21}$$

then

$$\frac{1}{NM_0} \sum_{x \in \mathbb{Z}} \sum_{m \leq M_0} \prod_{i=0}^{\ell} \mathbf{1}_A(x + P_i(m)) \geq \exp\bigl(-\exp(C\delta^{-C})\bigr). \tag{22}$$

Proof. We spell out the deduction because Theorem 1.2 of [16] produces one popular parameter, whereas we need a lower bound averaged over all positive parameters. Set $\varepsilon = c_1\delta^{\ell+1}$, where $c_1 = c_1(\mathbf{P}) > 0$ is sufficiently small. After enlarging C, the fixed polynomials have $(C,1)$-coefficients. Since $\varepsilon = c_1\delta^{\ell+1}$, the size condition (21), with C chosen sufficiently large in terms of $\mathbf{P}$, excludes the alternative $N \leq \exp\bigl(\exp(O_{\mathbf{P}}(\varepsilon^{-O_d(1)}))\bigr)$ in [16, Proposition 5.1].[1] That proposition, applied with $f_0 = \cdots = f_\ell = \mathbf{1}_A$, therefore gives positive integers q, M_1 satisfying

$$M_0 \geq M_1 \geq M_0 \exp\bigl(-\exp(C\delta^{-C})\bigr), \qquad q \leq \exp\bigl(\exp(C\delta^{-C})\bigr), \tag{23}$$

and a local factor chain $\mathcal{B}_1, \dots, \mathcal{B}_\ell$ of modulus q and resolutions $M_1^{\deg P_1}, \dots, M_1^{\deg P_\ell}$ such that

$$\left| \frac{1}{N} \sum_x \mathbb{E}_{\substack{m \leq \varepsilon M_1 \\ q \mid m}} \prod_{i=0}^{\ell} \mathbf{1}_A(x + P_i(m)) - \frac{1}{N} \sum_x \mathbb{E}_{\substack{m \leq \varepsilon M_1 \\ q \mid m}} \mathbf{1}_A(x) \prod_{i=1}^{\ell} (\Pi_{\mathcal{B}_i} \mathbf{1}_A)(x + P_i(m)) \right| \leq \varepsilon. \tag{24}$$

Here $\Pi_{\mathcal{B}_i} f$ denotes the average of f on each atom of $\mathcal{B}_i$. If $m \leq \varepsilon M_1$ and $q \mid m$, then

$$q \mid P_i(m), \qquad |P_i(m)| \ll_{\mathbf{P}} \varepsilon M_1^{\deg P_i},$$

since $P_i(0) = 0$. Each atom of $\mathcal{B}_i$ is an arithmetic progression of step q and length comparable to $M_1^{\deg P_i}$. Thus the number of $x \in [N]$ for which x and $x + P_i(m)$ lie in different atoms is at most

$$\ll_{\mathbf{P}} \left(\frac{N}{M_1^{\deg P_i}} + 1 \right) \frac{|P_i(m)|}{q} \ll_{\mathbf{P}} \varepsilon N.$$

Consequently,

$$\frac{1}{N} \sum_x \mathbb{E}_{\substack{m \leq \varepsilon M_1 \\ q \mid m}} \mathbf{1}_A(x) \prod_{i=1}^{\ell} (\Pi_{\mathcal{B}_i} \mathbf{1}_A)(x + P_i(m)) = \frac{1}{N} \sum_x \mathbf{1}_A(x) \prod_{i=1}^{\ell} (\Pi_{\mathcal{B}_i} \mathbf{1}_A)(x) + O_{\mathbf{P}}(\varepsilon).$$

Since $\mathcal{B}_1$ refines every $\mathcal{B}_i$, the product on the right is $\mathcal{B}_1$-measurable. Hence

$$\frac{1}{N} \sum_x \mathbf{1}_A(x) \prod_{i=1}^{\ell} (\Pi_{\mathcal{B}_i} \mathbf{1}_A)(x) = \frac{1}{N} \sum_x (\Pi_{\mathcal{B}_1} \mathbf{1}_A(x))^2 \prod_{i=2}^{\ell} (\Pi_{\mathcal{B}_i} \mathbf{1}_A)(x).$$

So [16, Lemma 5.2], applied to the chain $\mathcal{B}_1, \mathcal{B}_1, \mathcal{B}_2, \dots, \mathcal{B}_\ell$, shows that this is at least $\delta^{\ell+1}$. Combining this with (24) and taking c_1 sufficiently small gives

$$\frac{1}{N} \sum_x \mathbb{E}_{\substack{m \leq \varepsilon M_1 \\ q \mid m}} \prod_{i=0}^{\ell} \mathbf{1}_A(x + P_i(m)) \geq \delta^{\ell+1} - O_{\mathbf{P}}(\varepsilon) \geq \tfrac{1}{2}\delta^{\ell+1}. \tag{25}$$

The size assumption (21), with C enlarged if necessary, ensures $\varepsilon M_1/q \geq 2$, and

$$|\{m \leq \varepsilon M_1 : q \mid m\}| = \left\lfloor \frac{\varepsilon M_1}{q} \right\rfloor \geq \frac{\varepsilon M_1}{2q}.$$

---

[1] We use Proposition 5.1 of [16] with the evident correction

$$M \geq N^{1/d} \exp\bigl(-\exp(O(\varepsilon^{-O(1)}))\bigr).$$

The displayed lower bound in the published statement has misplaced parentheses and is incompatible with the accompanying bound $M \leq N^{1/d}$.

Since $\varepsilon M_1 \le M_0$ and all summands are nonnegative, opening (25) gives

$$\begin{aligned}
&\frac{1}{NM_0}\sum_{x}\sum_{m\le M_0}\prod_{i=0}^{\ell}\mathbf{1}_A(x+P_i(m))\\
&\ge \frac{|\{m\le \varepsilon M_1 : q \mid m\}|}{M_0}\left(\frac{1}{N}\sum_{x}\mathbb{E}_{\substack{m\le \varepsilon M_1\\ q\mid m}}\prod_{i=0}^{\ell}\mathbf{1}_A(x+P_i(m))\right)\\
&\ge \frac{\varepsilon\delta^{\ell+1}}{4q}\frac{M_1}{M_0}.
\end{aligned}$$

Using (23) and $\varepsilon = c_1\delta^{\ell+1}$, the last expression is at least

$$\exp\bigl(-\exp(C\delta^{-C})\bigr)$$

after enlarging C. This proves (22). □

Proof of Theorem 1.1. Put $M = \lfloor N^{1/d}\rfloor$, $\gamma = 1/c$, and

$$F_A(x,m) := \prod_{i=0}^{\ell}\mathbf{1}_A(x+P_i(m)).$$

Strict monotonicity of $n\mapsto\lfloor n^c\rfloor$ gives the exact reindexing identity

$$\mathcal{C}_{c,\mathbf{P}}(A;M) = \sum_{x}\sum_{m\le M}\chi_c(m)F_A(x,m). \tag{26}$$

Since $N \asymp_d M^d$, Proposition 2.5 and (11) give

$$\frac{\mathcal{C}_{c,\mathbf{P}}(A;M)}{NM^{\gamma}} = \frac{1}{NM^{\gamma}}\sum_{x,m\le M}\Delta_c(m)F_A(x,m) + O_{c,\mathbf{P}}(M^{1-\gamma-\beta}). \tag{27}$$

The function $m\mapsto\Delta_c(m)$ is positive and decreasing, and the mean value theorem gives

$$\Delta_c(M) \asymp_c M^{\gamma-1}. \tag{28}$$

Consequently Lemma 2.6 implies

$$\begin{aligned}
\frac{1}{NM^{\gamma}}\sum_{x,m\le M}\Delta_c(m)F_A(x,m) &\ge \frac{\Delta_c(M)}{M^{\gamma-1}}\left(\frac{1}{NM}\sum_{x,m\le M}F_A(x,m)\right)\\
&\gg_c \exp\bigl(-\exp(C\delta^{-C})\bigr).
\end{aligned}$$

By (19), the error in (27) is $O(M^{-\beta/2})$. Increasing C to absorb the fixed positive constant and taking $\zeta = \beta/2$ proves (3).

Now assume $\delta \ge (\log\log N)^{-\kappa}$. Choose $\kappa > 0$ so small that $\kappa C < 1/2$. Then

$$\exp(C\delta^{-C}) \le \exp\bigl(C(\log\log N)^{\kappa C}\bigr) = o(\log N),$$

so the positive term in (3) is $N^{-o(1)}$. It dominates the fixed power $M^{-\zeta}$ for all sufficiently large N, and the count is positive.

Finally, for every $1\le i\le\ell$, the polynomial P_i is nonzero, and, for every $1\le i<j\le\ell$, the polynomial $P_i - P_j$ is nonzero because P_i and P_j have distinct degrees. Consequently, the equations

$$P_i(m) = 0 \quad\text{and}\quad P_i(m) = P_j(m)$$

have only finitely many positive integer solutions. The contribution of all such m to (26) is $O_{\mathbf{P}}(N)$, but the preceding lower bound is $NM^{\gamma}N^{-o(1)}$. Removing these parameters leaves a pairwise distinct configuration. □

3. Pointwise ergodic averages

The reference pointwise theorem is [10, Theorem 1.3]: for integer polynomials of distinct degrees, the averages (5) converge pointwise almost everywhere. It remains to show that sampling the parameter along $\mathcal{S}_c$ does not change their limit.

We first record the elementary summability fact needed for the smooth model.

Lemma 3.1 (Riesz means)**.** *Let (z_m) be a bounded complex sequence and suppose*

$$\frac{1}{M}\sum_{m\le M} z_m \longrightarrow z.$$

For $0<\gamma<1$, put $\Delta(m)=(m+1)^\gamma - m^\gamma$ and $D_M=\sum_{m\le M}\Delta(m)$. Then

$$\frac{1}{D_M}\sum_{m\le M}\Delta(m)z_m \longrightarrow z. \tag{29}$$

Proof. Put $a_k := \frac{1}{k}\sum_{m\le k} z_m$, so that $a_k\to z$. Since $0<\gamma<1$, the sequence $\Delta(m)$ is positive and decreasing. Summation by parts gives

$$\frac{1}{D_M}\sum_{m\le M}\Delta(m)z_m = \sum_{k<M}\frac{k(\Delta(k)-\Delta(k+1))}{D_M}a_k + \frac{M\Delta(M)}{D_M}a_M.$$

The coefficients are nonnegative, tend to zero for each fixed k, and, by telescoping, have sum

$$\frac{\sum_{k<M} k(\Delta(k)-\Delta(k+1)) + M\Delta(M)}{D_M} = 1.$$

Thus they satisfy the hypotheses of the Toeplitz theorem [5, Theorem 2.6], which proves (29). □

Proof of Theorem 1.2. By homogeneity suppose $\|f_i\|_\infty\le 1$, and write

$$z_m(x) := \prod_{i=1}^{\ell} f_i(T^{P_i(m)}x), \qquad M_R := \lfloor R^c\rfloor, \qquad \gamma = 1/c.$$

Reindexing by $m=\lfloor n^c\rfloor$ gives

$$A_R^{(c)}(\mathbf{f}) = \frac{1}{R}\sum_{m\le M_R}\chi_c(m)z_m. \tag{30}$$

Using (11), (19), and (20),

$$\left\| A_R^{(c)}(\mathbf{f}) - \frac{1}{R}\sum_{m\le M_R}\Delta_c(m)z_m\right\|_1 \ll_{c,\mathbf{P}} \frac{M_R}{R}M_R^{-\beta} \ll_{c,\mathbf{P}} M_R^{1-\gamma-\beta} \ll_{c,\mathbf{P}} R^{-c\beta/2}. \tag{31}$$

By [10, Theorem 1.3], for almost every x the averages $M^{-1}\sum_{m\le M} z_m(x)$ converge to a limit $L(x)$. Lemma 3.1, applied pointwise, gives

$$\frac{1}{(M+1)^\gamma - 1}\sum_{m\le M}\Delta_c(m)z_m(x) \longrightarrow L(x). \tag{32}$$

Moreover,

$$(M_R+1)^\gamma - 1 = R + O(1). \tag{33}$$

Combining (32), with $M=M_R$, and (33), we obtain

$$\frac{1}{R}\sum_{m\le M_R}\Delta_c(m)z_m(x) = \frac{(M_R+1)^\gamma-1}{R}\left(\frac{1}{(M_R+1)^\gamma-1}\sum_{m\le M_R}\Delta_c(m)z_m(x)\right)\longrightarrow L(x)$$

for almost every x.

Choose $A>1+\frac{2}{c\beta}$ and put $R_j=\lfloor j^A\rfloor$. By (31),

$$\sum_{j=1}^{\infty}\left\| A_{R_j}^{(c)}(\mathbf{f}) - \frac{1}{R_j}\sum_{m\le M_{R_j}}\Delta_c(m)z_m\right\|_1 \ll_{c,\mathbf{P}} \sum_{j=1}^{\infty} R_j^{-c\beta/2} \ll_A \sum_{j=1}^{\infty} j^{-Ac\beta/2} < \infty.$$

Consequently, by Tonelli's theorem,

$$\int_X \sum_{j=1}^{\infty} \left| A_{R_j}^{(c)}(\mathbf{f})(x) - \frac{1}{R_j} \sum_{m \le M_{R_j}} \Delta_c(m) z_m(x) \right| d\nu(x) < \infty.$$

Thus, for almost every x,

$$\left| A_{R_j}^{(c)}(\mathbf{f})(x) - \frac{1}{R_j} \sum_{m \le M_{R_j}} \Delta_c(m) z_m(x) \right| \longrightarrow 0.$$

Since the second term converges almost everywhere to $L(x)$, it follows that

$$A_{R_j}^{(c)}(\mathbf{f})(x) \longrightarrow L(x)$$

for almost every x.

It remains to pass from (R_j) to all positive integers. If $R_j \le R < R_{j+1}$, then

$$\begin{aligned} A_R^{(c)}(\mathbf{f})(x) - A_{R_j}^{(c)}(\mathbf{f})(x) = &\left(\frac{1}{R} - \frac{1}{R_j}\right) \sum_{n \le R_j} \prod_{i=1}^{\ell} f_i\big(T^{P_i(\lfloor n^c \rfloor)} x\big) \\ &+ \frac{1}{R} \sum_{R_j < n \le R} \prod_{i=1}^{\ell} f_i\big(T^{P_i(\lfloor n^c \rfloor)} x\big). \end{aligned}$$

Since $\|f_i\|_\infty \le 1$, taking absolute values gives

$$\left| A_R^{(c)}(\mathbf{f})(x) - A_{R_j}^{(c)}(\mathbf{f})(x) \right| \le \left(\frac{1}{R_j} - \frac{1}{R}\right) R_j + \frac{R - R_j}{R} = 2\frac{R - R_j}{R} \le 2\frac{R_{j+1} - R_j}{R_j}.$$

Moreover, the mean value theorem gives

$$R_{j+1} - R_j \le (j+1)^A - j^A + 1 \ll_A j^{A-1}, \qquad R_j \gg_A j^A.$$

Hence

$$\sup_{R_j \le R < R_{j+1}} \left| A_R^{(c)}(\mathbf{f})(x) - A_{R_j}^{(c)}(\mathbf{f})(x) \right| \ll_A j^{-1} \longrightarrow 0$$

uniformly in x. This fills the gaps and proves convergence along the full sequence, with the same limit as (5). □

References

[1] V. Bergelson and A. Leibman, *Polynomial extensions of van der Waerden's and Szemerédi's theorems*, J. Amer. Math. Soc. **9** (1996), 725–753.

[2] M. Boshernitzan and M. Wierdl, *Ergodic theorems along sequences and Hardy fields*, Proc. Natl. Acad. Sci. USA **93** (1996), 8205–8207.

[3] T. H. Chan, A. V. Kumchev and M. Wierdl, *Additive bases arising from functions in a Hardy field*, Acta Math. Hungar. **129** (2010), no. 3, 263–276.

[4] L. Daskalakis, *Ergodic theorems for bilinear averages, Roth's theorem and corners along fractional powers*, arXiv:2504.18307.

[5] L. Daskalakis, Pointwise ergodic theorems for non-conventional bilinear averages along $(\lfloor n^c \rfloor, -\lfloor n^c \rfloor)$, *arXiv preprint arXiv:2503.03976*, 2025.

[6] J. Fornal and B. Krause, Double recurrence and almost sure convergence: primes and weighted theory, arXiv:2603.22203.

[7] N. Frantzikinakis and M. Wierdl, A Hardy field extension of Szemerédi's theorem, *Adv. Math.* **222** (2009), no. 1, 1–43.

[8] N. Frantzikinakis, *Multiple recurrence and convergence for Hardy sequences of polynomial growth*, J. Anal. Math. **112** (2010), 79–135.

[9] H. Furstenberg, *Ergodic behavior of diagonal measures and a theorem of Szemerédi on arithmetic progressions*, J. Analyse Math. **31** (1977), 204–256.

[10] D. Kosz, M. Mirek, S. Peluse, R. Wan and J. Wright, *The multilinear circle method and a question of Bergelson*, Ann. of Math., to appear; arXiv:2411.09478.

[11] M. O'Keeffe, Sárközy's theorem for fractional monomials, arXiv:2411.07386.

[12] S. Peluse, *Bounds for sets with no polynomial progressions*, Forum Math. Pi **8** (2020), Paper No. e16, 55 pp.

[13] J. Rivat and A. Sárközy, A sequence analog of the Piatetski–Shapiro problem, *Acta Math. Hungar.* **74** (1997), no. 3, 245–260.

[14] A. Sárközy, *On difference sets of sequences of integers. I*, Acta Math. Acad. Sci. Hungar. **31** (1978), no. 1–2, 125–149.

[15] A. Sárközy, *On difference sets of sequences of integers. III*, Acta Math. Acad. Sci. Hungar. **31** (1978), no. 3–4, 355–386.
[16] X. Shao and M. Wang, *Quantitative bounds in a popular polynomial Szemerédi theorem*, Proc. Roy. Soc. Edinburgh Sect. A, published online 2025, doi:10.1017/prm.2025.10103; arXiv:2505.16822v2.
[17] J. Teräväinen, *Pointwise convergence of ergodic averages with Möbius weight*, arXiv:2401.03174v2.
[18] J. D. Vaaler, *Some extremal functions in Fourier analysis*, Bull. Amer. Math. Soc. (N.S.) **12** (1985), 183–216.

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